ABSOLUTE / UPAJ

Crop-health AI made usable in the field

Connecting crop-image analysis, agronomy review and farmer-facing diagnosis and advice.

Role: Product leadership across the Absolute portfolio | Period: October 2021–November 2024

Reported outcomes and scope

500K+ · Reported crop-image dataset

14 crops · Reported diagnosis coverage

Diagnosis → advisory

The documented farmer-facing journey links model outputs to a next action.

The product problem

Smallholder farmers need a useful next step when a crop shows signs of disease. An image classification is insufficient if a farmer cannot capture the image reliably, understand the output or turn it into an appropriate action.

My role

I led product work around crop-disease detection and advisory, including the relationship between the dataset, agronomist review and the farmer journey. Engineering, modelling and domain expertise were part of delivering that capability.

What I built

The documented data scope is more than 500,000 crop images across 14 crops. The Professional Profile describes a labelling process and agronomist review loop to support ground truth.

The experience includes capturing a leaf image in low-connectivity conditions, processing it after reconnection and presenting diagnosis with actionable advisory. Device and inference constraints form part of the delivery problem, along with the advice itself.

The work before a farmer could receive an answer

The data scope was more than 500,000 crop images across fourteen crops. The project included image curation, labelling and an agronomist review loop. Domain review was part of preparing the product input, with specialists helping establish the meaning of the training labels.

My role was product leadership around the crop-disease capability and its farmer-facing delivery, working with engineering, modelling and agronomy expertise. The model and data work had to connect with the actual help-seeking flow: what evidence the farmer could capture, what the system could return and how that answer reached the user.

From taking a photograph to knowing the next step

The flow let a farmer photograph a leaf without connectivity and run the diagnosis when the phone reconnected. This separated the moment of evidence capture from the availability of a network. The diagnosis was delivered inside Upaj alongside advisory, rather than ending at a model label.

Crop-specific field playbooks and an OpenAI API-powered conversational layer were intended to turn the result into practical guidance. Engineering work also considered low-end phones and efficient inference. Those constraints belonged in delivery because a diagnosis product had to remain usable at the moment and place the farmer needed help.

GrowStac and the current advisory proposition

The October 2026 GrowStac public page describes AI/ML advisory delivered as an Upaj checklist, supported by farm data and agronomy services. It is a current visual reference for turning agricultural information into guidance. The historical disease-detection project establishes crop and dataset coverage; ownership of the present GrowStac page or every current advisory offering is not claimed.

What changed

The product scope covers fourteen crops and more than 500,000 images, with a farmer-facing diagnosis and advisory experience. That is the evidenced dataset and delivery scale used in this study. Yield improvement was described as potential in the earlier CV; later causal performance figures are not used as delivered outcomes.

The lesson from this work

The product pattern is translating a technical prediction into a field decision. Input quality, uncertainty, comprehensible advice and the user’s device conditions matter as much as presenting a model label.

Visual references and walkthroughs

Upaj: Growstac advisory

Upaj: Growstac advisory Public reference captured on 13 October 2026.

Upaj: Growstac technology

Upaj: Growstac technology Public reference captured on 13 October 2026.

Documented workflow: Farmer image capture → Compression and reconnect → Crop-health diagnosis → Agronomist-grounded guidance → Farmer advisory and action

An authored explanation of the workflow described in this case study. This is a diagram, not a captured product screen.

View the online case study for the full image gallery and playable videos.

Evidence and attribution

Dataset and crop coverage are source-reported. Later precision, yield and support-deflection claims are not treated as verified outcomes.

Later yield, model-precision and support-deflection claims lack attached evaluations. Language counts vary across versions. Current GrowStac descriptions are a public reference after the historical employment period.